

Ex. No.: 11b)

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LRU

Aim:

To write a c program to implement LRU page replacement algorithm.

Algorithm:

- 1: Start the process
- 2: Declare the size
- 3: Get the number of pages to be inserted
- 4: Get the value
- 5: Declare counter and stack
- 6: Select the least recently used page by counter value 7:
- Stack them according the selection.
- 8: Display the values
- 9: Stop the process **Program**

Code:

```
#include <stdio.h>

int findLRU(int time[], int n) {    int
i, minimum = time[0], pos = 0;
    for (i = 1; i < n; ++i) {
if (time[i] < minimum) {
minimum = time[i];
pos = i;
        }    }
return pos;
}

int main() {
    int frames[10], pages[30], time[10], counter = 0, faults = 0;
    int n, f, i, j, flag1, flag2, pos;

    printf("Enter number of frames: ");
    scanf("%d", &f);

    printf("Enter number of pages: ");
    scanf("%d", &n);

    printf("Enter reference string: ");
    for (i = 0; i < n; ++i) {
        scanf("%d", &pages[i]);
    }
```

```

    for (i = 0; i < f; ++i) {
frames[i] = -1;    time[i]
= 0;
    }

    printf("\n");

    for (i = 0; i < n; ++i) {
        flag1 = flag2 = 0;

        for (j = 0; j < f; ++j) {
if (frames[j] == pages[i]) {
counter++;    time[j] =
counter;    flag1 = flag2 =
1;    break;
        }
    }

    if (flag1 == 0) {    for
(j = 0; j < f; ++j) {    if
(frames[j] == -1) {
counter++;
faults++;    frames[j]
= pages[i];    time[j] =
counter;    flag2 = 1;
break;
        }
    }
}

    if (flag2 == 0) {
pos = findLRU(time, f);
counter++;    faults++;
frames[pos] = pages[i];
    time[pos] = counter;
    }

    for (j = 0; j < f; ++j) {
if (frames[j] != -1)
printf("%d ", frames[j]);
    else
printf("-1 ");
    }
    printf("\n");
}

printf("\nTotal Page Faults = %d\n", faults);

```

```
return 0; }
```

Sample Output :

```
Enter number of frames: 3
Enter number of pages: 6
Enter reference string: 5 7 5 6 7 3
5 -1 -1
5 7 -1
5 7 -1
5 7 6
5 7 6 3
7 6
Total Page Faults = 4
```

Result:

Thus the algorithm is executed successfully.